Amir Carmel, Ph.D.
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Research Summary

My research is in algorithms for massive data: coresets and data summarization, sublinear and streaming algorithms, and metric embeddings, with a focus on clustering. At the Weizmann Institute, together with Robert Krauthgamer, I introduced *stable coresets*, a guarantee between weak and strong coresets, and showed that plain uniform sampling achieves it with size independent of the number of points. This yields new coresets for ℓ_1 and related metrics, with applications to fair clustering and k-median approximation (ICLR 2026), and optimal stable coresets for the geometric median (APPROX 2026). Related work studies coresets for 1-center in ℓ_1 metrics (ITCS 2025, TALG 2026) and coresets for clustering from noisy comparisons with few distance queries (NeurIPS 2026). At Penn State I worked on streaming algorithms for fitting tree metrics and ultrametrics (ICALP 2025) and on weighted rank aggregation (ICALP 2026). My PhD was in combinatorial pattern matching and computational biology.

Work Experience

2025–present **Postdoctoral Researcher**, *Weizmann Institute of Science*, Rehovot, Israel, Host: Robert Krauthgamer.

2024–2025 **Postdoctoral Researcher**, *Pennsylvania State University*, Pennsylvania, United States, Host: Debarati Das.

2024 **Postdoctoral Researcher**, *Weizmann Institute of Science*, Rehovot, Israel, Host: Robert Krauthgamer.

2021–2023 **Senior Data Scientist**, *Pagaya*.

- Participated in researching and improving the core business model of Pagaya in the auto loans industry.
- Studied and implemented a novel survival model for discrete time & competing risks, and analyzed its benefits on mortgage market.

2019–2021 **Algorithm Developer & System Engineer**, *Israeli Navy*, IDF.
Designed and implemented new sophisticated algorithms based on big data as a hands-on team leader.

2017–2019 **Senior Software Engineer**, *Israeli Navy*, IDF.
Brought a prototype to production level, while maintaining the current Navy's C4I system. Designed and incorporated the new system algorithmic pipelines.

2015–2017 **Software Engineer**, *Israeli Navy*, IDF.
Took a part in a small R&D team that designed and implemented a highly distributed C4I system.

2013–2018 **Teaching Assistant & Private tutor**, *Ben Gurion University & Technion*.
Courses: Combinatorial Algorithms, Algorithm 2, Design of Algorithms, and Algebra courses.

Education

2017–2021 **Master of Science**, *Technion*, Pure Mathematics, *Avg. – 91.0*.

2015–2019 **Doctor of Philosophy**, *Ben Gurion University*, Computer Science.

2013–2014 **Master of Science**, *Ben Gurion University*, Computer Science, *Avg. – 98.48*.

2011–2013 **Bachelor of Science**, *Ben Gurion University*, Math & Computer Science, *Avg. – 89.27*.

Ph.D. & M.Sc. Experience

2013 – 2018 **M.Sc & Ph.D at Ben Gurion University of the Negev**, *title: "Combinatorial Algorithms with Applications to Bioinformatics"*, Utilizing string algorithms and graph theory to design faster and more space-efficient algorithms in the domain of comparative genomics.

Advisors : Prof. Michal Ziv-Ukelson and Prof. Dekel Tsur, *Computer Science Department*

2019 – 2021 **M.Sc at Technion – Israel Institute of Technology**, *title: "On the odd area of discs and arcs"*, Studied a well known open problem that seeks to determine the minimum possible area of all the points in the plane that belong to an odd number of unit discs.

Advisor : Prof. Rom Pinchasi, *Faculty of Mathematics*

2014 **Summer Internship at Moscow Institute of Physics and Technology**, *title: "Partial Fragment Chaining"*, Studied the problem of chaining fragments in multiple sequence alignment, allowing cases where fragments may not appear in all genome sequences.

Advisor : Prof. Mikhail A. Roytberg (R.I.P), *Lab of Applied Mathematics*

Awards & Fellowships

2025 The Sir Charles Clore Postdoctoral Prize, Weizmann Institute of Science

2018 Research Excellence Award, Department of Computer Science, Ben-Gurion University, for the paper "On Almost Monge All Scores Matrices" (*Algorithmica*)

2015 Research Excellence Award, Department of Computer Science, Ben-Gurion University, for the paper "The Worst Case Complexity of Maximum Parsimony" (*Journal of Computational Biology*)

Peer-reviewed Publications

2026 Amir Carmel, Robert Krauthgamer, and Nir Petruschka. **Optimal Stable Coresets for Geometric Median via Uniform Sampling**. In *Approximation, Randomization, and Combinatorial Optimization. Algorithms and Techniques (APPROX/RANDOM 2026)*, volume 392 of *LIPIcs*, pages 30:1–30:14. Schloss Dagstuhl – Leibniz-Zentrum für Informatik, 2026.

2026 Amir Carmel and Robert Krauthgamer. **Stable Coresets: Unleashing the Power of Uniform Sampling**. In *14th International Conference on Learning Representations (ICLR 2026)*, 2026.

2026 Amir Carmel and Robert Krauthgamer. **Coresets for Clustering Using Noisy Comparisons and Few Distance Queries**. In *Advances in Neural Information Processing Systems 39 (NeurIPS 2026)*, 2026. To appear.

2026 Amir Carmel, Chengzhi Guo, Shaofeng H.-C. Jiang, and Robert Krauthgamer. **Coresets for 1-Center in ℓ_1 Metrics**. *ACM Transactions on Algorithms*, 2026. Conditionally accepted.

2026 Amir Carmel, Debarati Das, and Tien-Long Nguyen. A scalable and unified framework to weighted rank aggregation. In *53rd International Colloquium on Automata, Languages, and Programming (ICALP 2026)*, volume 374 of *LIPIcs*, pages 49:1–49:23. Schloss Dagstuhl – Leibniz-Zentrum für Informatik, 2026.

2025 Amir Carmel, Chengzhi Guo, Shaofeng H.-C. Jiang, and Robert Krauthgamer. Coresets for 1-center in ℓ_1 metrics. In *16th Innovations in Theoretical Computer Science Conference (ITCS 2025)*, volume 325 of *LIPIcs*, pages 28:1–28:20. Schloss Dagstuhl – Leibniz-Zentrum für Informatik, 2025.

2025 Amir Carmel, Debarati Das, Evangelos Kipouridis, and Evangelos Pipis. Fitting tree metrics and ultrametrics in data streams. In *52nd International Colloquium on Automata, Languages, and Programming (ICALP 2025)*, volume 334 of *LIPIcs*, pages 42:1–42:21. Schloss Dagstuhl – Leibniz-Zentrum für Informatik, 2025.

2023 Amir Carmel and Rom Pinchasi. Some notes about the odd area of unit discs centered at points on a circle. *AUSTRALASIAN JOURNAL OF COMBINATORICS*, volume 87, pages 148–159, 2023.

2019 Amir Carmel, Dekel Tsur, and Michal Ziv-Ukelson. On almost monge all scores matrices. *Algorithmica*, volume 81, pages 47–68. Springer, 2019.

2016 Amir Carmel, Dekel Tsur, and Michal Ziv-Ukelson. On almost Monge all scores matrices. In *27th Annual Symposium on Combinatorial Pattern Matching (CPM 2016)*, volume 54 of *LIPIcs*, pages 17:1–17:12. Schloss Dagstuhl – Leibniz-Zentrum für Informatik, 2016.

2016 Amir Carmel, Yonat Shemer-Avni, Ilan Smoly, Esti Yeger-Lotem, and Michal Ziv-Ukelson. Algorithms for regular tree grammar network search and their application to mining human–viral infection patterns. *Journal of Computational Biology*, volume 23, pages 165–179. Mary Ann Liebert, Inc. 140 Huguenot Street, 3rd Floor New Rochelle, NY 10801 USA, 2016.

2015 Amir Carmel, Yonat Shemer-Avni, Ilan Smoly, Esti Yeger-Lotem, and Michal Ziv-Ukelson. Algorithms for regular tree grammar network search and their application to mining human-viral infection patterns. In *15th International Workshop on Algorithms in Bioinformatics (WABI 2015)*, volume 9289 of *LNCS*, pages 53–65. Springer, 2015.

2014 Amir Carmel, Noa Musa-Lempel, Dekel Tsur, and Michal Ziv-Ukelson. The worst case complexity of maximum parsimony. In *25th Annual Symposium on Combinatorial Pattern Matching (CPM 2014)*, volume 8486 of *LNCS*, pages 79–88. Springer, 2014.

2014 Amir Carmel, Noa Musa-Lempel, Dekel Tsur, and Michal Ziv-Ukelson. The worst case complexity of maximum parsimony. *Journal of Computational Biology*, volume 21, pages 799–808. Mary Ann Liebert, Inc. 140 Huguenot Street, 3rd Floor New Rochelle, NY 10801 USA, 2014.

Papers Under Submission

2026 Amir Carmel, Robert Krauthgamer, and Nir Petruschka. Terminal dimensionality reduction in ℓ_1, 2026.

2026 Amir Carmel and Robert Krauthgamer. Coresets for k-means: When does uniform sampling suffice?, 2026.

2026 Amir Carmel, Yakov Kosoburd, and Robert Krauthgamer. Recovery beats storage: Improved space for preprocessed 3SUM, 2026. arXiv:2608.22355, Submitted to STOC 2027.